

Fall 2025 - CS 228: Biometric Security with AI

Assignment 1: Gait-Based User Identification and Verification

Due Date: Oct 15th, 11:59PM

AI assistance instruction: You may use AI assistance for this assignment but make sure to understand and explain your thought process. Develop code in modules or blocks and double check for correctness and logical soundness. You must be ready to explain your approach/code/submission clearly when asked.

1. Objective

The goal of this assignment is to provide hands-on experience in designing, implementing, and evaluating a simple biometric security systems using some basic neural networks. You will work with a real-world gait dataset collected from mobile phone sensors. The dataset is shared alongside this assignment as “sample_data.zip”. You will implement two distinct systems: **user identification** and **user verification**, using two different neural network architectures: a **Feedforward Neural Network (FNN)** with feature engineering and a **Long Short-Term Memory (LSTM)** network on raw time-series data.

This assignment will test your understanding of:

- Time-series data preprocessing (windowing, outlier detection, smoothing).
- Feature extraction and dimensionality reduction (feature subset selection).
- Design and implementation of FNN and LSTM models for classification tasks.
- Model evaluation using cross-validation and appropriate performance metrics.
- Comparative analysis of different modeling approaches for biometric systems.

2. Dataset Description

The provided dataset contains gait data from **10 distinct users**. For each user, there are two separate data files:

1. **Gyroscope:** Contains 3-axis rotational velocity data (X, Y, Z).
2. **Accelerometer:** Contains 3-axis acceleration data (X, Y, Z).

You will need to combine the data from both sensors to form a single 6-dimensional time-series signal for each user.

3. Assignment Tasks

This assignment is divided into four main parts: data preprocessing, user identification experiments, user verification experiments, and a final report.

Part 1: Data Preprocessing and Preparation

Before training any models, you must process the raw sensor data into a usable format.

1. **Data Consolidation:** For each user, combine the accelerometer and gyroscope data into a single data structure or file. The resulting signal for each user should have 6 channels (Acc-X, Acc-Y, Acc-Z, Gyro-X, Gyro-Y, Gyro-Z).
2. **Windowing:** Apply a sliding window approach to segment the continuous time-series data into fixed-size samples.
 - **Window Size:** 5 seconds.

- **Overlap:** 50%.
 - Each resulting window will be a single sample for your models. You will need to determine the number of data points in a 5-second window based on the sensor's sampling rate (assume a standard rate if not provided, e.g., 50Hz or 100Hz, and state your assumption).
 - You can limit per user samples to around ~150 samples.
3. **Data Cleaning & Analysis:** Raw sensor data is often noisy. You must analyze the data within the windows for each user and apply at least one cleaning technique.
- **Analysis:** Visualize the signals for few different users. Do you observe noise or potential outliers?
 - **Cleaning:** Choose a suitable approach for **outlier detection** (e.g., Z-score, Interquartile Range) and/or **smoothing** (e.g., Moving Average, Savitzky-Golay filter).
 - **Justification:** In your final report, you must justify why you chose a particular method based on your analysis of the data. Apply the chosen technique(s) to all windows.

Part 2: Experiment I - User Identification (1-in-N Classification)

In this experiment, you will build a system that can identify a user from the group of 10 known users. This is a multi-class classification problem.

2.1. Feedforward Neural Network (FNN) Approach

1. **Feature Extraction:** From each 6-channel window, extract **at least 20 different features**. These can be statistical, frequency-domain, or other types of features. Examples include:
 - *Statistical:* Mean, Standard Deviation, Variance, Skewness, Kurtosis, Min, Max, Range, Interquartile Range.
 - *Frequency-Domain:* Dominant frequency components from a Fast Fourier Transform (FFT), spectral energy.
 - *Other:* Zero-Crossing Rate, Mean Absolute Deviation. Calculate these features for each of the 6 signal channels.
2. **Feature Subset Selection:** Use a suitable feature selection method to reduce the feature set from 20+ down to the **best 10 features**.
 - *Suggested Methods:* Principal Component Analysis (PCA), Correlation, SelectKBest (using metrics like ANOVA F-value or chi-squared), or Recursive Feature Elimination (RFE).
 - **Justify your choice of method** in your report.
3. **FNN Model Design:**
 - **Input Layer:** 10 neurons (one for each selected feature).
 - **Hidden Layers:** A maximum of 2 hidden layers, with a maximum of 5 neurons each. You can decide on the activation functions (e.g., ReLU, Tanh).
 - **Output Layer:** A layer suitable for 10-class classification (e.g., 10 neurons with a Softmax activation function).
 - **Data Scaling:** Remember to scale your features (e.g., using StandardScaler or MinMaxScaler) before feeding them to the network.

2.2. Long Short-Term Memory (LSTM) Approach

1. **Model Input:** For the LSTM, you will use the preprocessed (cleaned) windowed data directly, without feature extraction. The input for each sample will be a sequence of 6-dimensional data points.
2. **LSTM Model Design:**

- **Input Layer:** The input shape should correspond to the number of time steps in your window and the 6 features (channels).
- **Hidden Layers:** You can use a maximum of 5 hidden LSTM layers, each with a maximum of 6 neurons. You are free to decide on the connectivity (e.g., `return_sequences=True/False`).
- **Output Layer:** The final layer(s) should be suitable for 10-class classification (e.g., a Dense layer with Softmax activation).

Evaluation for Both Identification Approaches:

- **Cross-Validation:** You must use a cross-validation strategy (e.g., K-Fold or Stratified K-Fold) to train and evaluate your models.
- **Performance Metrics:** Report the average performance across the folds using:
 - Accuracy
 - Precision, Recall, and F1-Score (for each class and macro/weighted average).
 - Confusion Matrix.
- **Plots:** Provide plots showing the model's average and standard deviation of training and validation accuracy/loss over epochs.

Part 3: Experiment II - User Verification (1-vs-1 Authentication)

In this experiment, you will build a system that verifies a user's identity claim. This is a binary classification problem. You should frame this as a "one-vs-all" problem: for each user, you will build a separate model that determines if an incoming sample belongs to them (genuine) or to any of the other 9 users (impostor).

3.1. FNN Approach

1. **Data & Features:** You can reuse the 10 selected features from Part 2. For each of the 10 user-specific models, you will label the data as "genuine" (that user's data) or "impostor" (data from the other 9 users).
2. **FNN Model Design:**
 - **Input/Hidden Layers:** Same constraints as the identification task.
 - **Output Layer:** A single neuron with a Sigmoid activation function, suitable for binary classification.

3.2. LSTM Approach

1. **Data:** Reuse the cleaned, windowed signal data. Label it as "genuine"/1 or "impostor"/0 for each of the 10 models (does not have to be done parallelly).
2. **LSTM Model Design:**
 - **Input/Hidden Layers:** Same constraints as the identification task.
 - **Output Layer:** A single neuron with a Sigmoid activation function.

Evaluation for Both Verification Approaches:

- **Cross-Validation:** Use an appropriate cross-validation strategy. Be mindful of the class imbalance in the dataset for each model.
- **Performance Metrics:** Since accuracy can be misleading with imbalanced data, report the following metrics averaged over the 10 models:
 - False Acceptance Rate (FAR)
 - False Rejection Rate (FRR)

- Equal Error Rate (EER) - the point where FAR = FRR.
- Area Under the Curve (AUC) from the Receiver Operating Characteristic (ROC) curve.
- **Plots:** For one of the user models, provide the ROC curve plot.

Part 4: Report and Analysis (Maximum 3 pages, including plots, tables and observations)

You must submit a single, well-structured report in PDF format that includes the following sections:

1. **Introduction:** A brief overview of the methods you used.
2. **Methodology:**
 - **Preprocessing:** Detail your windowing and data cleaning choices with justifications.
 - **FNN Approach:** List the features you extracted, the selection method you used, the final 10 features, and your final network architecture.
 - **LSTM Approach:** Describe your final network architecture.
3. **Results:**
 - Present the performance metrics for both identification and verification experiments in clearly labeled tables.
 - Include all required plots (training curves, confusion matrix, ROC curve). Ensure plots are clearly titled and labeled.
4. **Discussion and Insights:**
 - Compare the performance of the FNN and LSTM approaches for both identification and verification. Which performed better and why do you think that is?
 - Compare the results of the identification vs. verification experiments.
 - Discuss any challenges you faced during the assignment (e.g., long training times, data imbalance, feature selection difficulties).
 - Provide any other insights or observations you made.
5. **Conclusion:** Briefly summarize your work and the key findings.

4. Submission Guidelines

- **What to Submit:**
 1. A single PDF report containing all the sections described in Part 4.
 2. A .zip file containing your well-commented source code (e.g., Python scripts or Jupyter notebooks).

5. Grading Rubric

Component	Points
Data Preprocessing & Preparation	15
Experiment I: Identification (FNN)	20
Experiment I: Identification (LSTM)	20
Experiment II: Verification (FNN)	20

Experiment II: Verification (LSTM)	20
Report Quality & Final Analysis	5
Total	100